

Kusuma Satya Sreeja Chalasani

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Professional Summary

Data Scientist and AI Engineer with experience building AI-powered applications, RAG systems, NLP solutions, and data-driven automation. Skilled in Python, SQL, machine learning, LLM-based application development, and cloud platforms. Experienced in developing end-to-end AI systems, data pipelines, and intelligent workflows that solve real-world operational challenges.

Skills

- **Programming Languages:** Python (Pandas, NumPy, Scikit-learn), SQL
- **Artificial Intelligence & Machine Learning:** Machine Learning (Regression, Classification, Clustering), Natural Language Processing (NLP), Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, Agentic AI Workflows, Vector Databases, Model Training & Evaluation, OpenAI API, Google Gemini API
- **Data & Analytics:** Data Cleaning, Feature Engineering, Exploratory Data Analysis (EDA), Data Transformation, Data Validation, Data Visualization, Tableau, Data Pipelines, Microsoft Excel (Advanced), Dashboard creation, PowerPoint
- **Web & Application Development:** React.js, TypeScript, Flask, REST API Development, Backend Development, Full-Stack Development, Tailwind CSS, Streamlit
- **Databases:** PostgreSQL, SQLite, IndexedDB
- **Cloud & Tools:** Microsoft Azure, Google Cloud Platform (GCP), GitHub, Git, Splunk, ServiceNow, Jupyter Notebook, Google Colab, Visual Studio Code (VS Code)

Work Experience

University of North Texas - Cross Functional IT Support Specialist

Dec 2024 – May 2026 | Denton, TX

- Managed and maintained operational datasets for 240+ university devices, ensuring data accuracy and system availability
- Analyzed recurring technical issues and generated data-driven insights, improving troubleshooting efficiency by ~20%
- Developed and maintained data tracking, reporting, and documentation workflows for IT operations
- Supported system deployment, updates, and maintenance, reducing downtime and improving operational reliability
- Collaborated with cross-functional IT teams to ensure system performance, scalability, and efficient issue resolution
- Structured and organized datasets to support reporting, analytics, and decision-making processes

Cloud4c - Cloud Support Engineer | Data Engineering & Operations | Client: DHL

June 2022 - July 2024 | India

- Built and maintained ETL data pipelines and backend data processing systems using Python and SQL, enabling scalable analytics and ML workflows
- Processed and analyzed large-scale datasets to identify trends, support predictive analytics, and improve data-driven decision-making
- Automated data workflows using Python and PowerShell, reducing manual effort, improving processing efficiency by ~30%
- Performed trend analysis on cloud storage and performance metrics to support capacity planning, infrastructure optimization
- Worked with Microsoft Azure and Google Cloud Platform (GCP) for data processing, storage, and cloud-based analytics
- Prepared and transformed structured datasets for machine learning and analytics use cases
- Collaborated with cross-functional teams to ensure data integrity, system reliability, and high availability

Selected Software & AI Projects

AI-Powered Data Scientist Agent & Explainable AI Dashboard

[GitHub](#) | *March 2026 – April 2026*

- Developed an AI-powered agent that automates end-to-end data analysis, feature engineering, model training, evaluation, and explainability workflows
- Implemented automated EDA, feature engineering, and model evaluation pipelines, reducing manual analysis time by ~40%
- Integrated Explainable AI (XAI) techniques to improve model interpretability and transparency
- Developed interactive dashboards and visual analytics solutions to generate business insights and support data-driven decision-making

Technologies: Python, Scikit-learn, Pandas, Streamlit, React, OpenAI API

RAG-Based AI Interview Prep Assistant

[GitHub](#) | *March 2026*

- Designed a Retrieval-Augmented Generation (RAG) system for document-based Q&A using dynamic PDF ingestion
- Implemented embedding generation and vector search (ChromaDB), improving response relevance by ~30%

- Built an interactive UI for querying documents with source-backed responses
 - Applied prompt engineering to reduce hallucinations and ensure context-aware answers
- Technologies:** Python, LangChain, ChromaDB, Streamlit, Sentence Transformers, Groq API

AI Guardian: Fraudulent Job Post Detection System

[GitHub](#) | *University of North Texas | Jan 2026 - April 2026*

- Developed an anomaly detection model (Isolation Forest) to identify fraudulent job postings from metadata features
- Performed exploratory data analysis (EDA) to identify fraud trends and patterns, enabling improved anomaly detection accuracy
- Built a hybrid ML model combining NLP (BERT) and anomaly detection for enhanced classification performance
- Evaluated model performance using precision, recall, and F1-score on imbalanced datasets, achieving strong detection accuracy

Technologies: Python, NLP (BERT), Isolation Forest, XGBoost, SHAP, Streamlit

Intelli AI – AI-Powered Internal Talent Matching System

Dallas AI Summer Program | June 2025 - Aug 2025

- Built an NLP-based job matching system to align candidate skills with job descriptions, achieving >90% matching accuracy
- Developed semantic matching workflows using NLP and similarity scoring to align employee skills with job requirements
- Automated email notifications using Gmail API for real-time communication
- Improved recommendation accuracy through iterative feature optimization

Technologies: Python, PostgreSQL, Gmail API, NLP, AI

Real-Time Crime Data Streaming & Visualization - Google Cloud Platform

University of North Texas | April 2025 - May 2025

- Built a real-time data pipeline using PySpark Structured Streaming on Google Cloud Dataproc
- Processed and analyzed large-scale streaming data to identify crime trends, generate actionable insights, and support predictive analytics workflows
- Performed data aggregation and visualization using BigQuery and Python

Technologies: PySpark, Google Cloud Platform (Dataproc, BigQuery, Cloud Storage), Python

LUMA: AI-Powered Emotional Wellness Companion

Sep 2025 – Oct 2025

- Developed a full-stack web application with frontend (React, TypeScript) and backend API integration for personalized AI-driven emotional support
- Built an emotion-aware conversational AI system with context retention and tone analysis
- Implemented mood tracking with AI-generated visualizations and journaling features
- Designed analytics dashboards to track user engagement and mood trends

Technologies: React, TypeScript, Google Gemini API, IndexedDB, Tailwind CSS, Recharts, Web Speech API

Education

Master of Science, Data Science | University of North Texas | *Aug 2024 - May 2026*

Bachelor of Science, Data Science and Data Analytics | Loyola Academy Degree & PG College | *2019 - 2022*

AI Programs & Fellowships

Dallas AI Summer Program | Southern Methodist University (SMU) | Summer 2025

- Selected for an 8-week applied AI program featuring mentorship from industry leaders across AI, cloud, and enterprise technology organizations.
- Collaborated on Intelli AI, an AI-powered internal talent matching prototype leveraging NLP, semantic similarity, and workflow automation.
- Worked under industry mentorship to design and develop practical AI solutions addressing real-world business challenges.
- Participated in technical workshops, project showcases, and collaborative problem-solving sessions focused on applied AI and machine learning.
- Presented project outcomes and technical approach during the program's final showcase.

Certifications

- Claude 101 [Certificate](#) | Anthropic | June 2026
- AI Fluency: Framework & Foundations [Certificate](#) | Anthropic | March 2026
- Introduction to Model Context Protocol (MCP) [Certificate](#) | Anthropic | March 2026
- Oracle Cloud Infrastructure Generative AI [Certificate](#) | Oracle | Oct 2025
- Azure Fundamentals AZ - 900 [Certificate](#) | Microsoft Certified | Feb 2023